

1. 5 Core Components of Data Communication

- 1. **Message:** Actual data payload to communicate (Text, Number, Audio, Video).
- 2. **Sender:** Workstation/phone generating the source bit stream.
- 3. **Receiver:** Sink device accepting the delivered information.
- 4. **Transmission Medium:** Guided (Twisted pair, Coax, Fiber) or Unguided (RF).
- 5. **Protocol:** Agreed rules governing Syntax, Semantics, and Timing.

Effectiveness: Delivery (Right node), Accuracy, Timeliness, Jitter (Delay variance)

2. Data Flow & Transmission Modes

Mode	Direction	Channel Allocation	Real-World Example
Simplex	Strictly 1-Way (Unidirectional)	100% capacity in 1 direction	Keyboard to CPU, TV broadcast
Half-Duplex	2-Way, but 1 at a time	Alternating time sharing	Walkie-Talkie ("Over & Out")
Full-Duplex	Simultaneous 2-Way	Divided / Separate channels	Telephone Call, Switched Ethernet

Connections: Point-to-Point (Dedicated link) vs Multipoint (Shared capacity among >2 nodes).

3. Physical Network Topologies Formula & Trade-off Matrix (for n Nodes)

Topology	Physical Links Formula	I/O Ports/Node	Cabling Cost	Fault Tolerance Profile	Key Failure Bottleneck
Mesh	$n(n - 1) / 2$	$n - 1$	Highest $O(n^2)$	Maximum; Dedicated alternate routes	Excessive wiring & physical port saturation
Star	n	1	Moderate $O(n)$	High (1 broken cable isolates only that host)	Central Hub / Switch Failure collapses entire network
Bus	1 backbone + n drops	1 tap	Lowest $O(1)$	Poor (Backbone cut halts entire bus)	Cable break or missing terminator stops all reflections
Ring	n (Unidirectional)	2 (In / Out)	Low to Moderate	Single point of failure (Dual ring FDDI fixes)	Token circulation break halts transmission
Tree / Hybrid	Hierarchical ($\$n-1\$$)	1 per leaf	Moderate to High	Isolated to sub-branch	Root hub / Trunk line failure disconnects branches

4. Geographical Network Categories

Category	Range	Typical Medium	Standard
PAN	< 10 m	Bluetooth, Zigbee, IrDA	IEEE 802.15
LAN	100 m - 2 km	Twisted Pair, Wi-Fi	IEEE 802.3, 802.11
MAN	5 - 50 km	Optical Fiber, Coaxial	Cable TV, Metro Eth
WAN	> 100 km	Submarine Fiber, Satellite	IP, BGP, MPLS

5. 3-Tier ISP Architecture & Internet Structure

- Tier 1 ISPs:** Global backbone carriers (AT&T, Lumen, Tata). Own trans-oceanic cables. *Zero transit fees*; free settlement peering.
- Tier 2 ISPs:** Regional operators (Airtel, Jio). Buy transit from Tier 1; peer at **IXPs (Internet Exchange Points)**.
- Tier 3 ISPs:** Local retail broadband (ACT, Hathway, FTTH). Last-mile access to homes and businesses.
- POP (Point of Presence):** Physical interface facility where external networks hook into an ISP backbone.

6. Protocols, Standards Organizations & Layering Principles

Protocol 3 Elements:

- Syntax:** Data structure & bit field layout.
- Semantics:** Meaning & control actions.
- Timing:** Speed matching & sequencing.

De Jure vs De Facto:

- De Jure:* Legally approved by treaty bodies (ISO 7498, IEEE 802.3).
- De Facto:* Established by industry adoption without formal law (TCP/IP).

Standards Organizations:

- ISO:** OSI Reference Model (7 Layers).
- IEEE:** LAN standards (802.3, 802.11).
- IETF:** Internet Standards via RFCs.
- ITU-T:** Telecommunications sector.

Service Access Points (SAPs): Layer (N-1) offers service to Layer N through SAPs using Primitives: Request, Indication, Response, Confirm.

1. OSI 7-Layer Reference Model (ISO 7498) Functional Deep Dive

#	Layer Name	PDU	Primary Core Functions & Protocols	Addressing & Hardware
7	Application	Data	Direct network interface for user software: File transfer (FTP), Web browsing (HTTP/HTTPS), Email (SMTP/POP3/IMAP), Name resolution (DNS), Dynamic config (DHCP).	URL / URI, Email IDs
6	Presentation	Data	Translation: Character encoding (ASCII, EBCDIC). Encryption/Decryption: Cryptographic security (SSL/TLS). Compression: Lossy & lossless payload reduction.	MIME Types
5	Session	Data	Dialog Control: Enforces simplex, half-duplex, or full-duplex session turns. Synchronization & Checkpoints: Inserts recovery checkpoints in large data streams.	Session IDs, Tokens
4	Transport	Segment / Datagram	"Heart of OSI": End-to-end process-to-process delivery. Segmentation & Reassembly, Port addressing, Connection-oriented (TCP) vs Connectionless (UDP), End-to-end flow & error control.	Port Numbers (16-bit: 0 - 65535)
3	Network	Packet	Source-to-destination Host-to-Host delivery across heterogeneous internetworks. Logical IP addressing, Routing algorithms (Dijkstra, Distance Vector), Packetizing & fragmentation.	Logical IP Address (32/128-bit) <i>Routers, L3 Switches</i>
2	Data Link	Frame	Hop-to-hop Node-to-Node frame delivery. Framing (Bit/Byte stuffing), Physical MAC addressing, Flow control, Error control (CRC Trailer), Shared medium access control (CSMA/CD).	Physical MAC Address (48-bit) <i>Bridges, L2 Switches, NIC</i>
1	Physical	Bits	Transmission of raw binary stream (0s and 1s) over physical media. Voltage levels, Bit duration, Line coding (Manchester/NRZ), Transmission mode (Simplex/Duplex), Pinouts (RJ-45).	Signal Voltages, Optical pulses <i>Repeaters, Hubs, Modems</i>

2. Encapsulation & Trailer T2 Anatomy

- **Encapsulation (Top-to-Bottom):** Har layer apna protocol header append karti hai (\$H_7 o H_6 o H_5 o H_4 o H_3 o H_2\$).
- **Unique DLL Trailer (\$T_2\$):** Data link layer akeli layer hai jo Header (\$H_2\$) ke sath **Trailer (\$T_2\$)** bhi append karti hai.
- \$T_2\$ carries **CRC Checksum** hardware bit-error detection ke liye!
- **Decapsulation (Bottom-to-Top):** Receiving host par headers inspect aur strip hote hain.

3. The 3 Distinct Delivery Scopes (AKTU Must-Know)

- **1. Hop-to-Hop (Data Link Layer):** Frame ko ek router/switch se agle adjacent router tak move karta hai using **MAC Addresses**. MAC changes at every hop!
- **2. Host-to-Host (Network Layer):** Packet ko source host computer se destination host tak deliver karta hai using **IP Addresses**. IP stays constant end-to-end!
- **3. Process-to-Process (Transport Layer):** Packet ko computer ke exact running application (e.g. Chrome tab) tak deliver karta hai using **Port Numbers**.

4. Master 10-Point Comparison: OSI Model vs TCP/IP Protocol Suite

Comparison Dimension	OSI Reference Model	TCP/IP Protocol Suite
Origin & Design Goal	ISO standard (1984); Conceptual theoretical reference model.	ARPANET / DoD (1970s); Practical implementation for working network.
Layer Architecture	Strictly 7 Layers .	4 Layers (Network Access, Internet, Transport, Application).
Protocol vs Service Separation	Clear, strict separation (Service, Interface, Protocol).	Loose separation; Layers are built around specific protocols.
Network Layer Service	Supports both Connection-Oriented & Connectionless .	Strictly Connectionless (Internet Protocol - IP datagrams).
Transport Layer Service	Strictly Connection-Oriented .	Supports both Connection-Oriented (TCP) & Connectionless (UDP) .
Session & Presentation Layers	Separate dedicated layers with distinct protocols.	No separate layers; Functions handled inside Application layer.
Protocol Independence	Protocols easily replaceable; Model developed before protocols.	Protocols tightly coupled; Hard to replace IP or TCP.
Global Commercial Dominance	Never gained widespread commercial adoption.	Universal standard powering the entire global Internet!

5. 4 Levels of Network Addressing in TCP/IP

- **Physical (MAC) Address:** 48-bit (6 Bytes). Hexadecimal format: 00:1A:2B:3C:4D:5E. First 24 bits = OUI (Vendor ID); Last 24 bits = NIC Serial. Burned into hardware.
- **Logical (IP) Address:** Universal host identifier across subnets. IPv4 = 32-bit (4 Bytes dotted-decimal 192.168.1.1); IPv6 = 128-bit (16 Bytes hex).
- **Port Address:** 16-bit integer (0 to 65,535). Well-known: 0-1023 (HTTP 80, HTTPS 443, DNS 53, SSH 22). Registered: 1024-49151. Dynamic: 49152-65535.
- **Specific / Application Address:** User-friendly URLs (www.aktu.ac.in) and emails. DNS resolves URL \$ o\$ IP.

Socket Address = IP Address : Port Number (e.g. 192.168.1.10 : 8080)

1. Transmission Impairments & Decibels

- Attenuation:** Loss of energy over distance. Solved by Amplifiers/Repeaters.

$dB = 10 \log_{10}(P_2 / P_1) \quad | \quad dB = 20 \log_{10}(V_2 / V_1)$

Power halved $\implies -3 \text{ dB}$. Multi-stage: $\text{dB}_{\text{total}} = \text{dB}_1 + \text{dB}_2$.
- Distortion:** Phase shift in composite signals (frequencies travel at different velocities).
- Noise:** Thermal (kTB $\times$), Induced, Crosstalk (wire coupling), Impulse (lightning spikes).

$SNR = P_{\text{signal}} / P_{\text{noise}} \quad | \quad SNR_{dB} = 10 \log_{10}(SNR)$

2. Nyquist vs Shannon Channel Capacity

- Nyquist (Noiseless Ideal Channel):**

$C = 2 \cdot B \cdot \log_2(M) \text{ bps} \quad | \quad \text{Baud} = 2 \cdot B$

B = Bandwidth (Hz), M = Discrete signal voltage levels.
- Shannon (Noisy Real-World Channel):**

$C = B \cdot \log_2(1 + SNR) \text{ bps}$ (Absolute Physical Ceiling!)

Linear SNR $= 10^{\{\text{SNR}_{dB} / 10\}}$. If $B=3000 \text{ Hz}$, $\text{SNR}_{dB}=31.62 \text{ dB}$ $\implies C \approx 31.6 \text{ kbps}$.
- Synthesis (Find M):** $2B \log_2(M) \leq C_{\text{Shannon}}$ $\implies M \leq \sqrt{1 + \text{SNR}}$.

3. Network Latency Breakdown (4 Delay Components) & Performance Metrics

- Transmission Delay:** $T_t = \frac{L}{B}$ (Packet Length L / Bandwidth B). Independent of distance!
 - Propagation Delay:** $T_p = \frac{d}{v}$ (Distance d / Wave velocity $v \approx 2 \times 10^8 \text{ m/s}$). Independent of packet size!
 - Queuing Delay (T_q):** Time waiting in intermediate router buffer queues.
 - Processing Delay (T_{proc}):** Header inspection, CRC verify, route lookup.
- $\text{Total Latency } T = T_t + T_p + T_q + T_{\text{proc}} \quad | \quad \text{RTT} \approx 2 \cdot T_p + T_t$
- $\text{BDP (Bandwidth-Delay Product)} = \text{Bandwidth} \cdot T_p \text{ (Bits in Flight)}$
- Throughput:** Actual measured delivery rate ($\leq \text{Bandwidth}$).
 - Goodput:** Application payload delivery rate (excludes headers).
 - Jitter:** Delay variance between successive arriving packets.

4. Digital Transmission & Line Coding Schemes (Bit Stream: 0 1 0 0 1 1 1 0)

Scheme	Rule for Bit '0'	Rule for Bit '1'	Self-Sync	DC Component
Polar NRZ-L	High Voltage (+V)	Low Voltage (-V)	Poor on runs	Moderate
Polar NRZ-I	No change (Stay at current level)	Inversion at start of interval	Syncs on 1s only	Moderate
Polar RZ	Negative (-V) returns to 0 V at mid-bit	Positive (+V) returns to 0 V at mid-bit	Good	None
Manchester (802.3)	High-to-Low transition (↓) at mid-bit	Low-to-High transition (↑) at mid-bit	Perfect (Every bit)	Zero (None)
Diff Manchester	Transition at START of bit interval	NO transition at start of bit interval	Perfect (Mid-bit)	Zero (None)
Bipolar AMI	Zero Voltage (0 V)	Alternating +V and -V pulses	Syncs on 1s only	Zero (None)

5. Switching Techniques Compared

- Circuit Switching:** Dedicated physical path. 3 Phases (Setup $\rightarrow$ Data $\rightarrow$ Teardown). Zero queuing delay, but bandwidth wasted during silence. (PSTN Telephone).
- Message Switching:** Entire message stored and forwarded at every node. Huge hard drive buffers required. High delay. (Telegraph).
- Packet Switching:** Small packets (MTU). Statistical multiplexing, concurrent pipelining.
 - Datagram (Connectionless):* Packets routed independently; can arrive out-of-order; robust. (IP).
 - Virtual Circuit (Connection-oriented):* Pre-planned path (SVC/PVC); in-order arrival; uses VCI. (ATM).

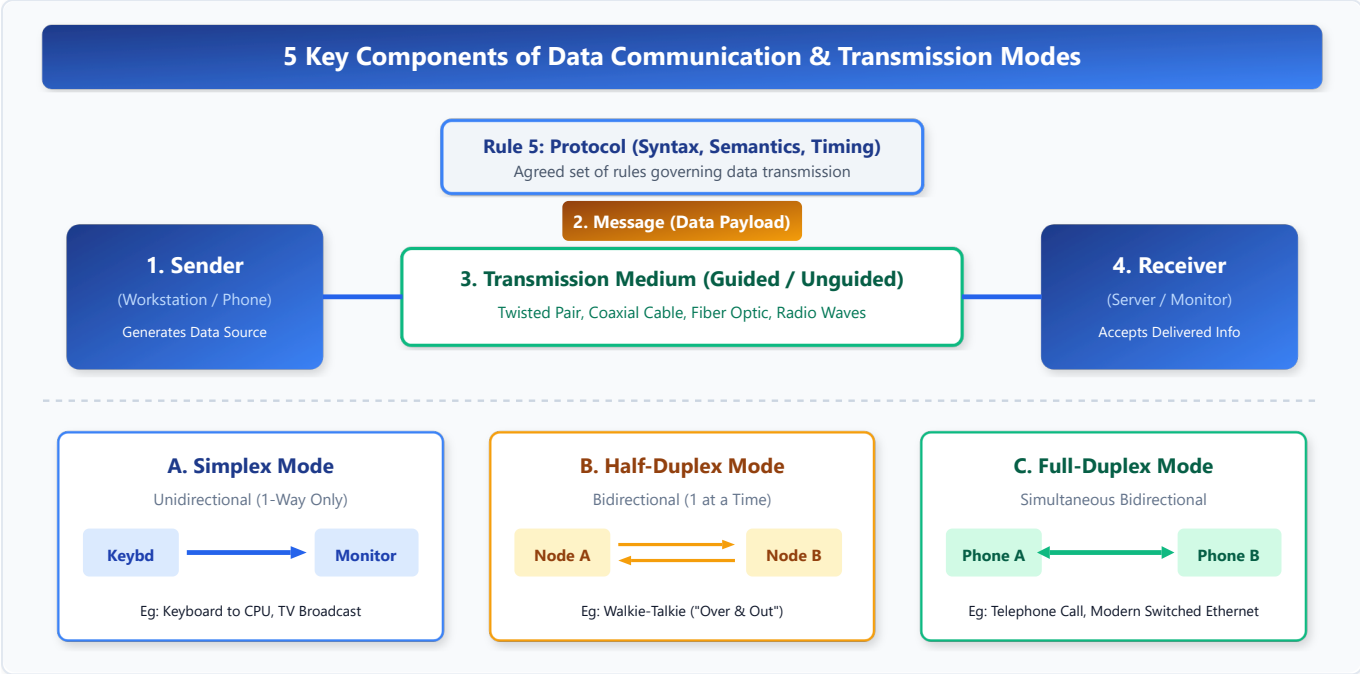
6. Network Devices & Domain Separation Matrix

Device	Layer	Collision Domain	Broadcast Domain
Repeater / Hub	L1 Physical	1 (Single shared pool)	1 (Floods all)
Bridge / Switch	L2 Data Link	N (1 per port)	1 (Floods broadcast)
Router	L3 Network	N (1 per port)	N (Blocks broadcast)
Gateway	L4-7 All	Separate	Separate

Golden Rule: Switches break Collision Domains. Routers break BOTH Collision and Broadcast Domains!

Module 01: Data Communication & Network Fundamentals

AKTU BCS603 / GATE CS Core Foundation: Data communication components, transmission modes, connection types, and network criteria.



- **Point-to-Point:** Dedicated physical link connecting exactly two devices. Link bandwidth 100% reserved. (e.g., Leased lines, satellite uplink/downlink).
- **Multipoint (Multidrop):** Single shared communication medium tapped into by multiple stations. Requires MAC arbitration protocols (CSMA/CD, Token passing) to prevent collisions.

AKTU Exam Tip: Jab bhi transmission mode par question aaye, Simplex, Half-Duplex, aur Full-Duplex ka directional diagram aur Walkie-Talkie vs Phone ka example zaroor draw karein!

Module 02: Network Topologies & Design Trade-offs

AKTU BCS603 Focus: Architectural layout, link calculation formulas, fault isolation, and comparative evaluation of Mesh, Star, Bus, Ring, and Hybrid topologies.

Physical Network Topologies: Geometric Layouts & Formulas

Mesh Topology

Links = $n(n-1)/2$
Ports/Node = $(n - 1)$

Star Topology

Links = n | Ports = 1
Central Single Point of Failure

Bus Topology

1 Backbone + n Drop Lines
Multipoint Shared Medium

Ring Topology

Links = n | Ports = 2
Token Ring (Unidirectional)

Topology	Physical Links	I/O Ports/Node	Cable Cost	Fault Tolerance	Key Bottleneck / Failure Point
Mesh	$n(n - 1) / 2$	$n - 1$	Very High	Maximum (Robust)	Excessive cabling & port requirement
Star	n	1	Moderate	Good (1 node fail ok)	Central Hub Failure collapses entire network
Bus	1 backbone + n drops	1 tap	Lowest	Poor	Backbone cable break stops all transmission
Ring	n (unidirectional)	2	Low to Moderate	Low (Dual ring helps)	Single break halts ring; token circulation delay
Tree/Hybrid	Hierarchical / Varies	Varies	Moderate to High	Isolated to segment	Root switch or trunk link failure isolates branch

1. Mathematical Formulation Summary (\$n\$ Nodes)

Topology	Link Formula	Ports per Device	Cabling Volume	Reliability Profile
Mesh	$n(n - 1) / 2$	$n - 1$	$O(n^2)$ — Maximum	Exceptional; Alternate paths exist for every node pair
Star	n	1	$O(n)$ — Moderate	High node resilience; Vulnerable to Central Hub failure
Bus	1 backbone + n drops	1 tap	$O(1)$ trunk — Lowest	Low; Backbone cut collapses whole segment
Ring	n	2 (In / Out)	$O(n)$ — Low	Moderate; Single node failure breaks token circulation

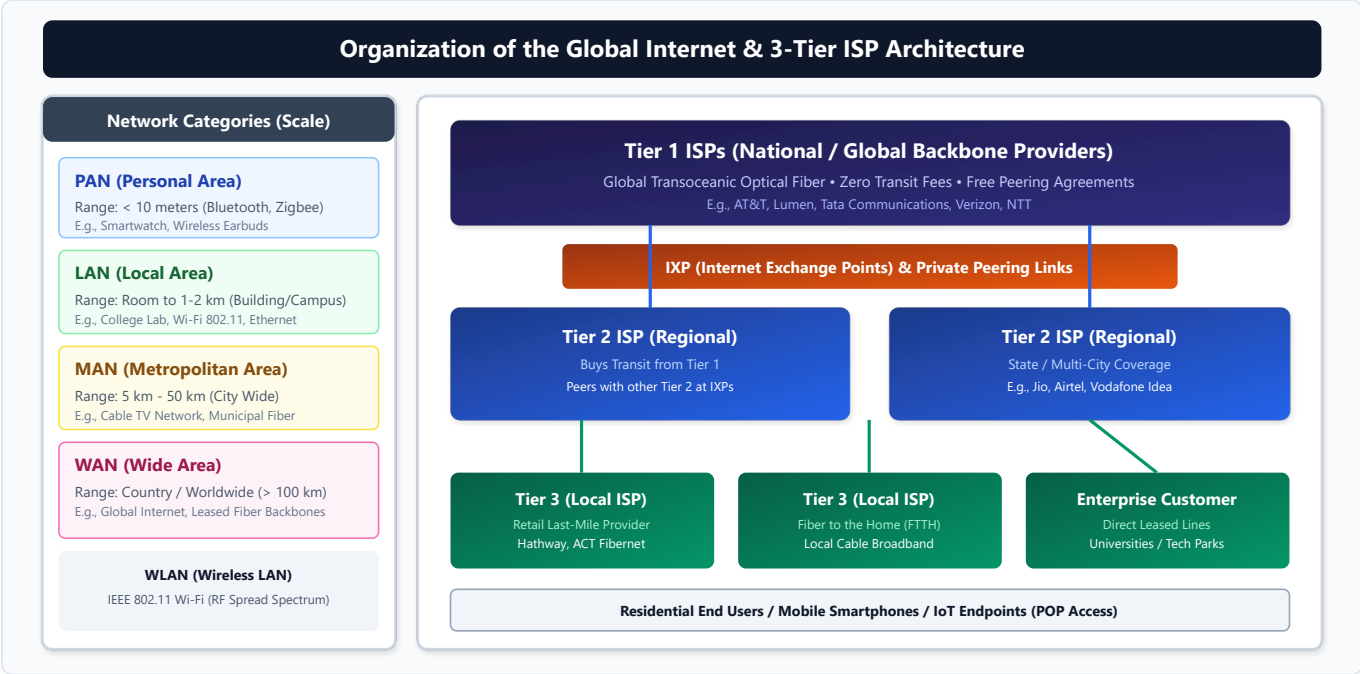
2. Star vs Mesh Comparison (Exam Favorite)

- Mesh Topology:** Provides guaranteed bandwidth and zero collision, but cable cost scales as $O(n^2)$. Practically deployed in core backbone networks (ISP Core, Defense).
- Star Topology:** Industry standard for Enterprise LANs (10/100/1000Base-T Ethernet). Cheap twisted pair cables, easy troubleshooting, plug-and-play expansion. Central failure risk is mitigated by using dual redundant Managed Switches.

AKTU Numerical Tip: Numerical solve karte waqt hamesha $n(n-1)/2$ formula likhein aur har step me values substitute karein: Agar $n=10$, Links = $(10 \times 9)/2 = 45$ cables!

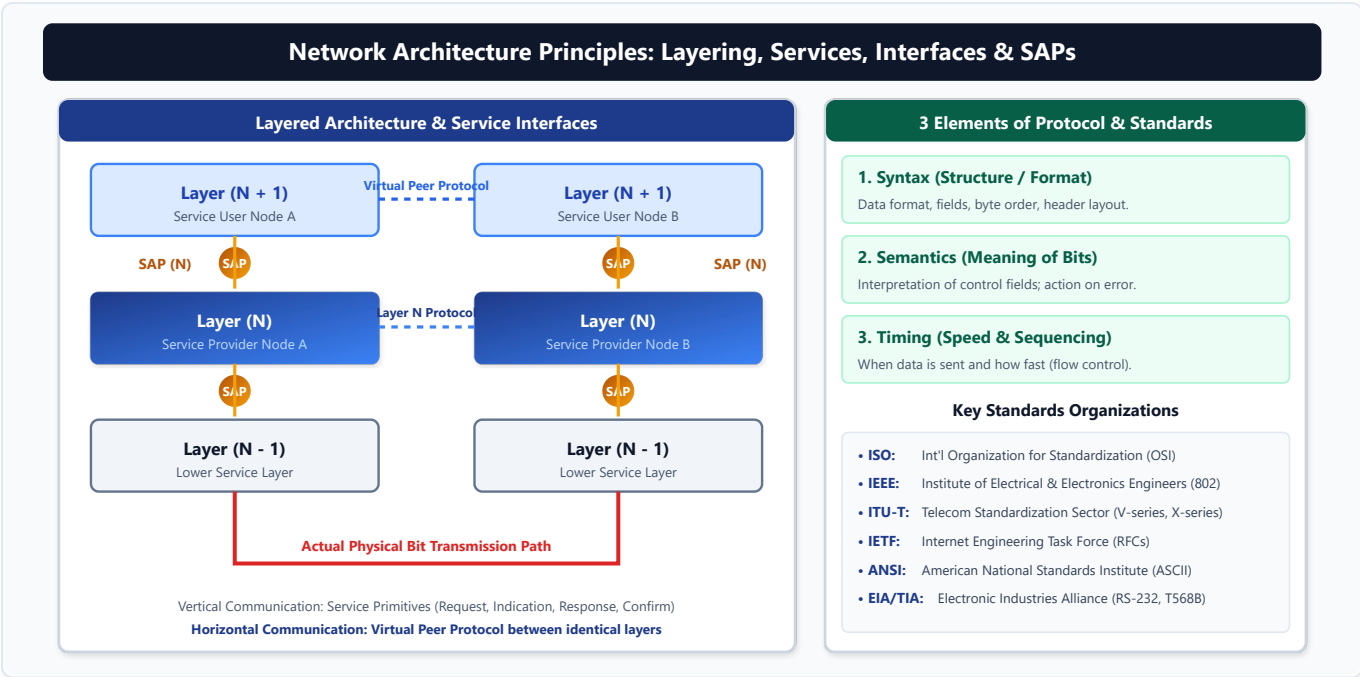
Module 03: Network Categories, Internet Structure & ISPs

AKTU BCS603 Focus: Distance-based classification (PAN, LAN, MAN, WAN), Hierarchical ISP structure (Tier 1, 2, 3), POPs, IXPs, and Internet governance.



Module 04: Protocols, Standards and Layering Principles

AKTU BCS603 Focus: Syntax, Semantics, Timing, De Facto vs De Jure, Standards bodies (ISO, IEEE, IETF, ITU-T), Layering concepts, SAPs, and Service Primitives.



1. 3 Key Pillars of Every Protocol

- **Syntax:** How data is structured into bit patterns (Header format, field offsets, flags).
- **Semantics:** Interpretation and actionable meaning assigned to bit patterns (e.g., error alert vs acknowledgment).
- **Timing:** Transmission synchronization, rate-matching flow control, and timeout thresholds.

2. Service vs Protocol Distinction

Dimension	Service (Vertical)	Protocol (Horizontal)
Interaction Axis	Between Layer (N) and Layer (N-1) on the <i>same</i> machine	Between Peer Layer (N) on Machine A and Peer Layer (N) on Machine B
Interface Point	Service Access Point (SAP) using Primitives (Request, Indication, Response, Confirm)	Network transmission medium carrying Protocol Data Units (PDUs)
Visibility	Internal to operating system stack / hardware drivers	Visible over wire as network packet headers

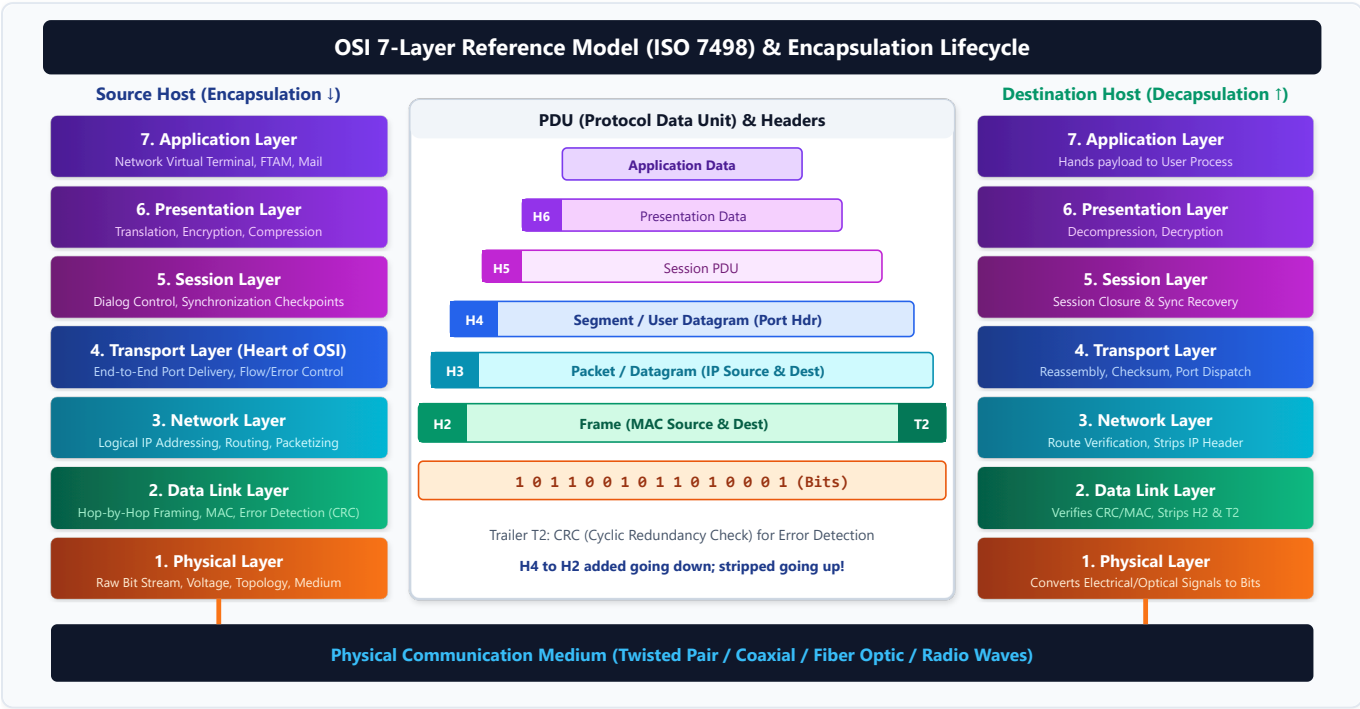
3. Standards Organizations Cheat Sheet

- **IEEE 802 Committee:** 802.3 (Ethernet), 802.11 (Wi-Fi), 802.15 (Bluetooth).
- **IETF (RFCs):** RFC 791 (IPv4), RFC 793 (TCP), RFC 2616 (HTTP/1.1).
- **ISO:** ISO 7498 Open Systems Interconnection (OSI Reference Model).

AKTU Critical Difference: De Facto means standards set by widespread custom without law (e.g. TCP/IP, USB Type-C popularity); De Jure means formal standards legalized by sanctioned treaty bodies (e.g. ISO 7498, IEEE specs).

Module 05: OSI Reference Model Architecture & 7 Layers

AKTU BCS603 Core Syllabus: ISO 7498 specification, functions of all 7 layers, PDU mapping, encapsulation headers, and trailer anatomy.



1. Master 7-Layer Summary Matrix

Layer # & Name	PDU	Primary Responsibility	Key Addressing / Identifiers	Associated Protocols / Hardware
7. Application	Data / Message	User UI & Network Service Access	URI / URL, Email IDs	HTTP, FTP, SMTP, DNS, SSH, SNMP
6. Presentation	Data	Data Translation, Encryption, Compression	MIME types, Character sets	SSL/TLS, ASCII, JPEG, MPEG, gzip
5. Session	Data	Dialog establishment, Checkpoints & Sync	Session IDs, Token controls	NetBIOS, PPTP, RPC, SIP
4. Transport	Segment / Datagram	End-to-End Process Delivery, Flow & Error	Port Numbers (16-bit: 0 - 65535)	TCP, UDP, SCTP
3. Network	Packet	Source-to-Destination Host Routing	Logical IP Address (32/128-bit)	IP, ICMP, OSPF, BGP <i>Routers</i>
2. Data Link	Frame	Hop-to-Hop Reliable Frame Delivery	Physical MAC Address (48-bit)	Ethernet 802.3, Wi-Fi 802.11 <i>Switches</i>
1. Physical	Bits	Transmission of raw binary stream	Voltages, RF carrier frequencies	EIA-232, RJ-45, Cat6, Fiber <i>Hubs</i>

2. Network Layer vs Transport Layer Delivery (Exam Favorite)

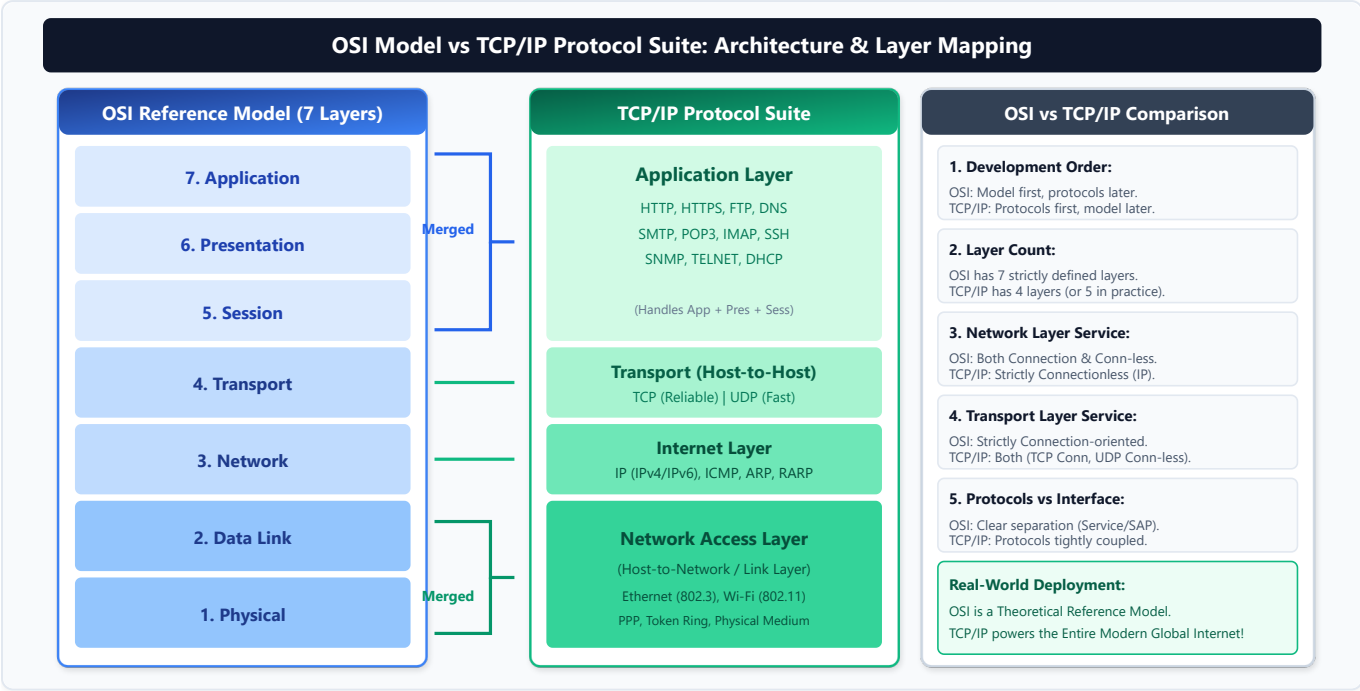
- Network Layer:** Sirf packet ko Source Host computer se Destination Host computer tak pahunchata hai (Host-to-Host delivery). Isse yeh nahi pata hota ki destination computer par kaunsa application software packet ka intezaar kar raha hai.

- **Transport Layer:** Packet ke andar ke Port Number ko inspect karke data ko destination computer ke exact running software process (e.g., Chrome tab vs Spotify) tak pahunchata hai (Process-to-Process delivery).

AKTU Exam Alert: "Describe all layers of the OSI model with a well-labeled diagram" 10-mark question me har layer ka PDU, 2-3 main functions, aur trailer \$T_2\$ ka role zaroor mention karein!

Module 06: TCP/IP Protocol Suite & OSI Comparison

AKTU BCS603 Focus: 4-Layer TCP/IP architecture, protocol stack (IP, ARP, ICMP, TCP, UDP), and authoritative 10-point OSI vs TCP/IP comparison matrix.



1. Core Protocols at Each TCP/IP Layer

- Application Layer:** HTTP/HTTPS (Web), FTP (File transfer), SMTP/IMAP (Email), DNS (Domain name resolution), DHCP (Dynamic IP assignment).
- Transport Layer:**
 - TCP:* Connection-oriented, reliable 3-way handshake, checksums, retransmission, flow control (Sliding window), congestion control.
 - UDP:* Connectionless, low overhead (8-byte header vs 20-byte TCP header), zero handshake latency, suitable for real-time media.
- Internet Layer:**
 - IP:* Best-effort, connectionless packet forwarding across autonomous networks.
 - ARP:* Resolves known IP address to unknown 48-bit MAC address via broadcast request and unicast reply.
 - ICMP:* Out-of-band network diagnostic signals (Echo Request / Echo Reply for ping).
- Network Access Layer:** IEEE 802.3 Ethernet framing, IEEE 802.11 Wi-Fi frames, physical transceivers.

2. OSI vs TCP/IP Architecture Comparison

Feature	OSI Reference Model	TCP/IP Model
Model Type	Conceptual & Descriptive	Functional & Implementation standard
Layer Count	7 Layers	4 Layers (RFC 1122) / 5 Layers
Network Layer	Connection-oriented and Connectionless	Strictly Connectionless (IP Datagram)
Transport Layer	Strictly Connection-oriented	Connection-oriented (TCP) & Connectionless (UDP)
Internet Adoption	Never gained widespread commercial adoption	The universal lingua franca of the global Internet

AKTU Exam Tip: Jab bhi TCP/IP vs OSI model ka question aaye, 4 vs 7 layer mapping diagram draw karein aur Transport vs Network layer ke connection modes (TCP/UDP vs IP) ka difference table me zaroor highlight karein!

Module 07: Addressing in Computer Networks

AKTU BCS603 Focus: 4 Addressing tiers (Physical MAC, Logical IP, Port, Specific), address translation protocols, and 3 delivery scopes (Hop-to-Hop vs Host-to-Host vs Process-to-Process).

4 Levels of Network Addressing & Delivery Scope

1. Physical Address (MAC / Link Layer)

- Length: 48 bits (6 Bytes) — Burned into NIC hardware
- Format: Hexadecimal

00:1A:2B:3C:4D:5E OUI (24 bits) Vendor ID NIC Specific ID (24 bits) • Scope: Hop-to-Hop (Node-to-Node) Delivery • Changes at every router hop across the transmission path! • Resolved by ARP (Address Resolution Protocol) 2. Logical Address (IP / Network Layer) • Length: IPv4 = 32 bits (4 Bytes) | IPv6 = 128 bits (16 Bytes) • IPv4 Format: Dotted Decimal 192.168.1.100 Network ID (NetID) Host ID (HostID) • Scope: Source-to-Destination (Host-to-Host) Delivery • REMAINS CONSTANT throughout entire end-to-end journey! • Assigned statically or dynamically via DHCP 3. Port Address (Transport Layer) • Length: 16 bits (0 to 65,535) • 3 Port Categories: 1. Well-known Ports (0 - 1023): HTTP=80, HTTPS=443, SSH=22 2. Registered Ports (1024 - 49151): MySQL=3306, Tomcat=8080 3. Dynamic / Private / Ephemeral (49152 - 65535): Client sockets • Scope: End-to-End (Process-to-Process) Delivery • Socket Address = IP Address : Port Number Identifies specific application process in OS 4. Specific / Application Address • Human-Friendly User-Level Addresses • Web URLs: <https://www.aktu.ac.in/index.html> • Email Address: student@aktu.edu.in • Fully Qualified Domain Names (FQDN) DNS translates Domain Name → 32-bit IP Address • Scope: User / Application Interface Layer Allows humans to navigate networks without memorizing numbers

1. Address Classification & Key Properties

Address Type	Bit Size	Hardware / OS Location	Delivery Horizon
Physical (MAC)	48 bits (6 Bytes)	NIC ROM / Firmware	Node-to-Node (Local subnet only)
Logical (IPv4)	32 bits (4 Bytes)	OS TCP/IP Stack Configuration	Host-to-Host (Global internetwork)
Logical (IPv6)	128 bits (16 Bytes)	OS TCP/IP Stack Configuration	Host-to-Host (Next-gen Internet)
Port Number	16 bits (0 - 65535)	Kernel Socket Table	Process-to-Process (Application binding)
URL / FQDN	Variable ASCII	Application layer / Browser cache	User interface (DNS resolved)

2. The Socket Abstraction

Network communication me process-to-process connection hamesha do **Sockets** ke beech banta hai:

Source Socket: 192.168.1.15 : 54321 (Client Ephemeral Port)

Destination Socket: 142.250.190.46 : 443 (Google Web Server HTTPS Port)

Golden Rule: Source & Destination IP addresses never change as packet moves from router to router. But Source & Destination MAC addresses change at EVERY hop!

Module 08: Transmission Impairments & Signal Characteristics

AKTU BCS603 Focus: Attenuation metrics (dB calculations), Distortion in composite signals, 4 types of noise, and Signal-to-Noise Ratio (SNR).

Transmission Impairments in Physical Media: Attenuation, Distortion & Noise

1. Attenuation (Loss of Energy)

Original Signal

Attenuated Signal

Amp (+dB)

$dB = 10 \log_{10}(P_2 / P_1)$

Negative dB = Loss | Positive dB = Gain

- Signal weakens over distance as heat
- Solved by: Amplifiers / Repeaters
- $dB = 20 \log_{10}(V_2 / V_1)$ for Voltages

2. Distortion (Change in Shape)

In Phase (At Sender)

Out of Phase (At Receiver)

Phase Shift & Delay Spread

Different frequencies travel at different speeds

- Occurs exclusively in composite signals
- Harmonic components arrive at different times
- Creates Inter-Symbol Interference (ISI)

3. Noise (External Corruption)

$SNR = \text{Signal Power} / \text{Noise Power}$ $SNR_{dB} = 10 \log_{10}(SNR)$

- Thermal Noise (Johnson-Nyquist kTB)
- Induced Noise (Motors, power cables)
- Crosstalk (Wire-pair interference)
- Impulse Noise (Spikes/Lightning - data killer)

Key Impairment Formulas & Engineering Solutions

Loss in Decibels: $dB = 10 \log_{10}(P_2 / P_1)$

If $P_2 = 0.5 P_1$, $dB = 10 \log_{10}(0.5) = -3 \text{ dB}$ (Half power loss)

Cascaded Decibels: $dB_{total} = dB_1 + dB_2 + dB_3$

Decibels simply ADD across multi-stage amplifiers and cable spans!

Signal-to-Noise Ratio: $SNR_{dB} = 10 \log_{10}(SNR)$

If $SNR = 1000$, $SNR_{dB} = 10 \log_{10}(1000) = 30 \text{ dB}$ (Clean transmission)

1. Comparative Impairment Diagnostic Matrix

Impairment	Primary Cause	Affected Signals	Engineering Remedy
Attenuation	Medium resistance & thermal absorption over distance	All Analog & Digital signals	Amplifiers (Analog) & Repeaters (Digital)
Distortion	Frequency-dependent propagation velocity (Phase shift)	Composite signals only	Equalizers & delay compensators
Thermal Noise	Random electron agitation (kTB)	All physical media	Cooling; Shielding; Shannon capacity bounds
Crosstalk	Mutual inductance between parallel copper wires	Twisted pair, multi-core cables	Differential signaling & tight wire twisting
Impulse Noise	Lightning strikes, power grid switches, sparking	Digital packets (Burst errors)	Shielding (STP), Error correcting codes (CRC/FEC)

2. Solved AKTU Numerical: Decibel Power Loss

Problem: Ek 10 km optical fiber link par input power $P_1 = 100 \text{ mW}$ hai. Receiver par received power $P_2 = 25 \text{ mW}$ measure hoti hai. Cable ka attenuation dB me aur dB/km me nikalein.

Solution:

$dB = 10 \log_{10}(\frac{25}{100}) = 10 \log_{10}(0.25) = 10 \times (-0.602) \approx -6.02 \text{ dB}$

Minus sign indicates power loss (Attenuation = 6.02 dB).

Per km attenuation $= \frac{6.02 \text{ dB}}{10 \text{ km}} = 0.602 \text{ dB/km}$.

Module 09: Channel Capacity Theorems: Nyquist & Shannon

AKTU BCS603 Numerical Core: Mathematical formulation of Nyquist bit rate, Shannon channel capacity, linear vs dB SNR conversions, and solved exam problems.

Data Rate Limits: Nyquist Bit Rate vs Shannon Capacity Theorem

1. Nyquist Bit Rate (Noiseless Channel)

$$C = 2 \cdot B \cdot \log_2(M) \text{ bps}$$

B = Bandwidth (Hertz) | M = Number of Discrete Signal Levels

- Assumption: Theoretical channel with ZERO noise.
- Baud Rate (Signal Rate) = $2 \cdot B$ signals/second.
- Bits carried per signal element = $\log_2(M)$ bits.
- Example: $B = 3000 \text{ Hz}$, Binary signal ($M = 2$):
 $C = 2 \cdot 3000 \cdot \log_2(2) = 6,000 \text{ bps} = 6 \text{ kbps}$
- If $M = 4$ levels (2 bits/level): $C = 12,000 \text{ bps} = 12 \text{ kbps}$
- If $M = 16$ levels (4 bits/level): $C = 24,000 \text{ bps} = 24 \text{ kbps}$

Notice: Nyquist allows infinite speed if $M \rightarrow \infty$ (Impossible in real world!)

2. Shannon Capacity (Noisy Real-World Channel)

$$C = B \cdot \log_2(1 + \text{SNR}) \text{ bps}$$

B = Bandwidth (Hertz) | SNR = Signal Power / Noise Power (Linear)

- Assumption: Real physical channel with Thermal Noise.
- Sets the ABSOLUTE THEORETICAL UPPER BOUND on data rate.
- Converting dB to Linear SNR:
 $\text{SNR}_{\text{dB}} = 10 \log_{10}(\text{SNR}) \Rightarrow \text{SNR} = 10^{(\text{SNR}_{\text{dB}} / 10)}$
- Real Telephone Line Example: $B = 3000 \text{ Hz}$, $\text{SNR}_{\text{dB}} = 31.62 \text{ dB}$:
 $\text{SNR} = 10^{3.162} \approx 1452 \Rightarrow \log_2(1453) \approx 10.5 \text{ bits}$
 $C = 3000 \cdot 10.5 \approx 31,500 \text{ bps} = 31.5 \text{ kbps}$ (V.34 Modem limit!)

Shannon capacity CANNOT be exceeded, no matter how many levels M are used!

Engineering Synthesis: How Engineers Combine Shannon & Nyquist to Find Signal Levels (M)

Step 1 (Upper Bound): Use Shannon's formula to find the maximum possible error-free channel capacity: $C_{\text{max}} = B \cdot \log_2(1 + \text{SNR})$

Step 2 (Signal Levels M): Equate Nyquist rate to Shannon Capacity: $2 \cdot B \cdot \log_2(M) = C_{\text{max}} \Rightarrow \log_2(M) = C_{\text{max}} / (2B) \Rightarrow M = 2^{(C_{\text{max}} / 2B)}$

Practical Lesson: Shannon tells us the fundamental physics ceiling; Nyquist tells us how many signal levels (constellation points) we must engineer!

1. Summary Comparison: Nyquist vs Shannon

Dimension	Nyquist Bit Rate	Shannon Channel Capacity
Formula	$C = 2 \cdot B \cdot \log_2(M) \text{ bps}$	$C = B \cdot \log_2(1 + \text{SNR}) \text{ bps}$
Channel Condition	Ideal Noiseless Channel	Realistic Thermal Noisy Channel
Primary Constraint	Number of discrete signal levels (\$M\$)	Thermal noise floor (SNR)
Physical Role	Tells how many signal levels must be encoded	Defines unbreakable speed ceiling set by nature

2. Solved Numerical: Combined Shannon & Nyquist

Question (AKTU Exam): Ek communication channel ki bandwidth \$B = 1 \text{ ext{ MHz}}\$ hai aur uska SNR \$63\$ hai. (a) Shannon capacity nikalein. (b) Nyquist formula ke mutabiq is data rate ko achieve karne ke liye kitne signal levels (\$M\$) chahiye?

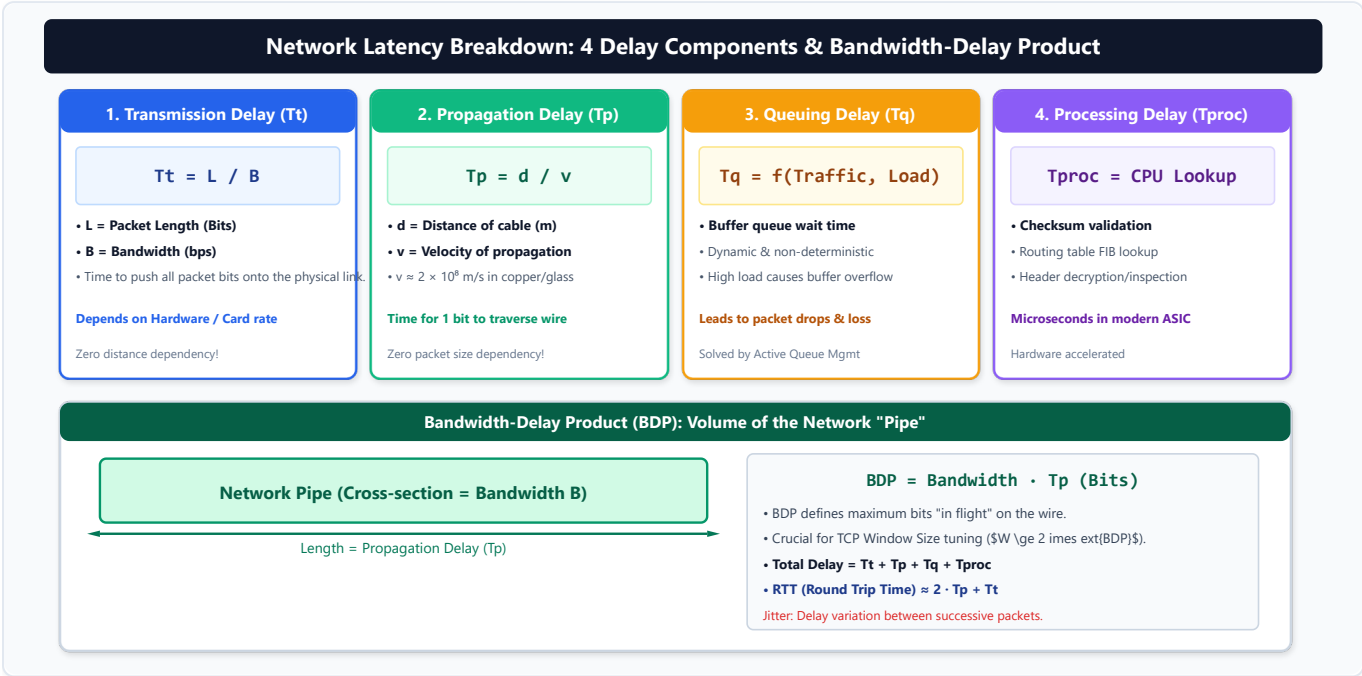
Solution:

(a) Shannon Capacity:
 $C = B \cdot \log_2(1 + 63) = 10^6 \cdot \log_2(64) = 10^6 \cdot 6 = 6 \text{ ext{ Mbps}}$

(b) Signal Levels (\$M\$):
 $C = 2 \cdot B \cdot \log_2(M) \implies 6 \cdot 10^6 = 2 \cdot 10^6 \cdot \log_2(M)$
 $\log_2(M) = \frac{6 \cdot 10^6}{2 \cdot 10^6} = 3 \implies M = 2^3 = 8 \text{ ext{ signal levels}}$

Module 10: Network Performance Metrics & Latency Analysis

AKTU BCS603 Focus: Latency decomposition (T_t , T_p , T_q , T_{proc}), Bandwidth-Delay Product (BDP), Round Trip Time (RTT), and jitter.



1. Transmission Delay vs Propagation Delay (AKTU Favorite)

Parameter	Transmission Delay (T_t)	Propagation Delay (T_p)
Core Formula	$T_t = L / B$	$T_p = d / v$
Governing Factors	Packet Size (L) & Link Bandwidth (B)	Cable Distance (d) & Wave Speed (v)
Analogous Meaning	Kitni der me pura packet car parking se nikla	Car ko highway par chalne me kitna time laga
Independent Of	Distance between nodes	Packet size and transmission bit rate

2. Channel Parameter Ratio α

Data link layer protocols ki efficiency ratio α par depend karti hai:

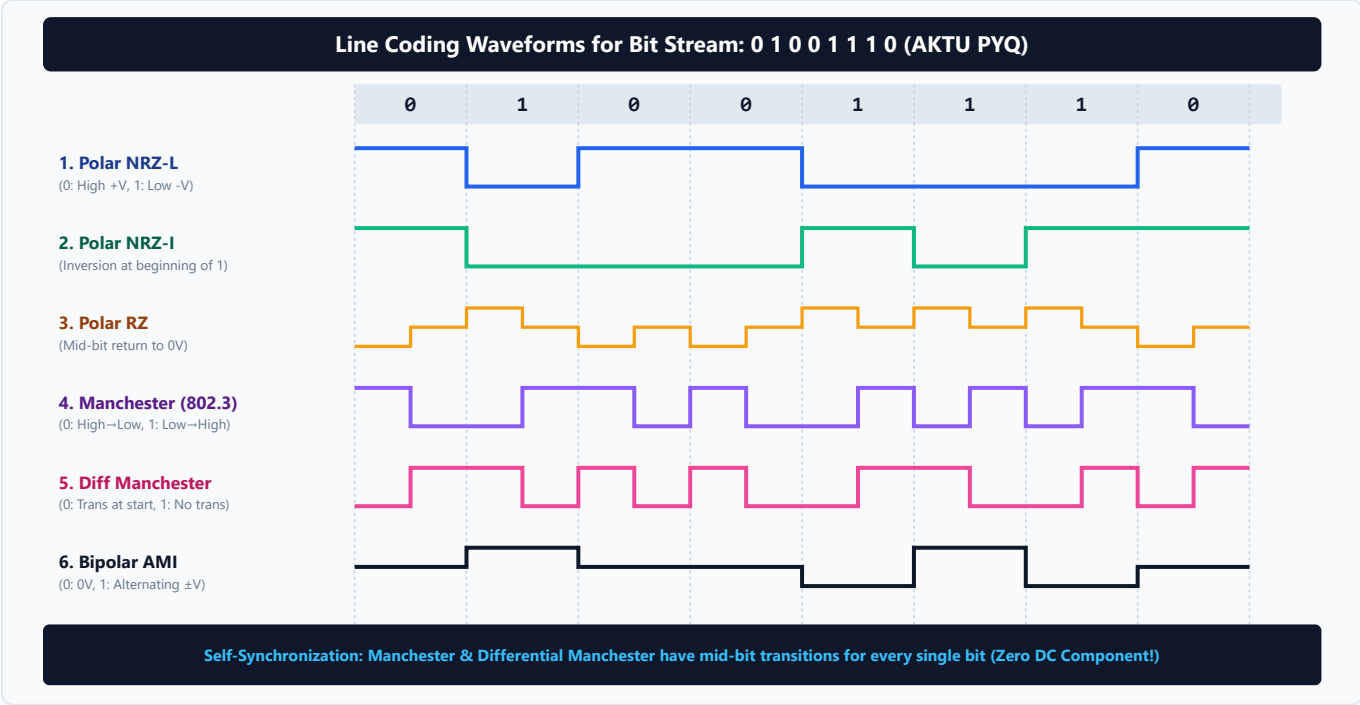
$$\alpha = \frac{T_p}{T_t} = \frac{d \cdot B}{v \cdot L}$$

Agar $\alpha \gg 1$ (High bandwidth, long distance jaise Satellite links), to Stop-and-Wait protocol extremely inefficient ho jata hai ($\approx \frac{1}{1 + 2\alpha} \rightarrow 0$).

AKTU Exam Tip: Numerical solve karte waqt bytes ko hamesha pehle bits me convert karein ($1 \text{ ext{ Byte}} = 8 \text{ ext{ bits}}$), aur Mega (10^6) ya Kilo (10^3) units ko standard SI units me convert karna na bhoolein!

Module 11: Digital Transmission & Line Coding Techniques

AKTU BCS603 Exam Mandatory: Comprehensive waveform generation, transition rules, DC component elimination, and comparative analysis of Line Coding techniques.



Module 12: Physical Transmission Media & Multiplexing

AKTU BCS603 Focus: Guided media specs, Optical Fiber TIR physics, Wireless frequency propagation, and Multiplexing architectures (FDM, WDM, Synchronous vs Statistical TDM).

Physical Transmission Media (Guided vs Unguided) & Multiplexing (FDM, WDM, TDM)

A. Guided Media (Wired Conduits)

1. Twisted Pair (UTP / STP):

- Twisting cancels electromagnetic interference (EMI) & crosstalk.
- Cat 5e (100 MHz, 1 Gbps), Cat 6 (250 MHz, 10 Gbps up to 55m). RJ-45.

2. Coaxial Cable:

- Center copper core, dielectric insulator, metallic braided shield, jacket.
- 50 Ω Baseband (digital), 75 Ω Broadband (analog cable TV). BNC connector.

3. Optical Fiber (Glass / Silica):

- Total Internal Reflection (TIR): Core index $n_1 >$ Cladding index n_2 .
- Single-mode (narrow core, laser, 100s km) vs Multi-mode (LED, short haul).
- Immense bandwidth, immune to EMI, zero spark hazard, lowest loss.**

B. Unguided Media (Wireless Propagation)

1. Radio Waves (3 kHz to 1 GHz):

- Omnidirectional antenna (broadcasts in all directions).
- Penetrates walls & buildings easily. FM Radio, AM, TV broadcast.

2. Terrestrial & Satellite Microwaves (1 to 300 GHz):

- Strictly Unidirectional line-of-sight propagation (dish antennas).
- Blocked by obstacles; requires repeater towers every 40-50 km.
- Geosynchronous Satellite uplink (6 GHz) & downlink (4 GHz).

3. Infrared (300 GHz to 400 THz):

- Strictly Short-range closed-room line-of-sight.
- Cannot penetrate walls (high security). TV remotes, wireless mice.

Multiplexing: Combining Multiple Signals onto a Single Physical Channel

FDM (Frequency Division)

- Analog signals divided into frequency bands.
- Separated by Guard Bands to avoid overlap.
- E.g., Radio broadcast, Cable TV channels.
- Channel bandwidth >> Signal bandwidth**

WDM (Wavelength Division)

- Optical equivalent of FDM for Fiber cables.
- Combines different wavelengths (colors of light).
- Multiplexer/Demultiplexer uses optical prism.
- DWDM delivers Terabits/sec over 1 fiber!**

TDM (Time Division)

- Digital signals share full bandwidth in time slots.
- Synchronous TDM: Pre-allocated time slots.
- Statistical TDM: On-demand dynamic slots.
- Eliminates wasted idle time slots!**

1. Single-Mode vs Multi-Mode Fiber Comparison

Feature	Single-Mode Fiber (SMF)	Multi-Mode Fiber (MMF)
Core Diameter	8 to 10 μm (Extremely thin)	50 to 62.5 μm (Wide core)
Light Source	Laser Diodes (Infrared 1310/1550 nm)	LEDs (Light Emitting Diodes 850 nm)
Modal Dispersion	Zero (Single ray travel)	High (Multiple rays bounce at different angles)
Distance Range	Up to 100+ km without amplification	Up to 2 km (Campus / Building backbones)

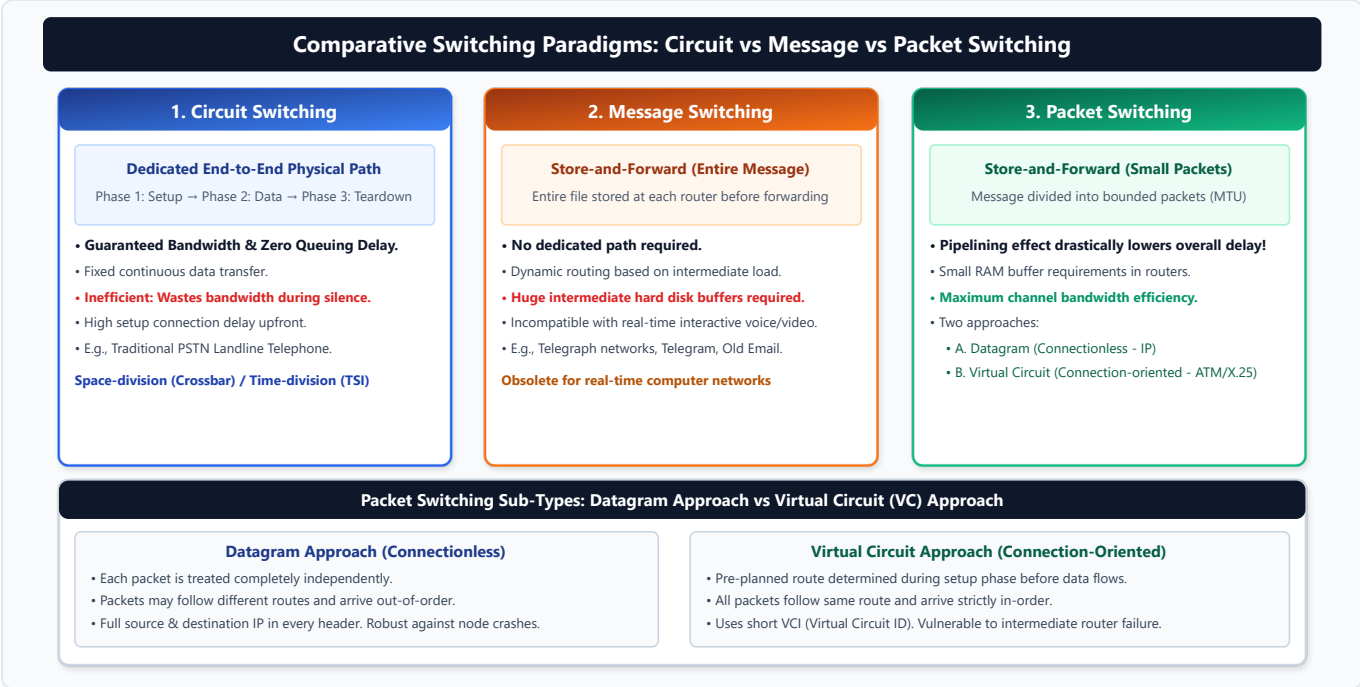
2. Synchronous TDM vs Statistical TDM

- Synchronous TDM:** Frame size is fixed. If host A is silent, slot A travels empty across the WAN. Efficiency drops when traffic is bursty.
- Statistical TDM:** Slots are dynamically addressed. Only stations with data in buffer are awarded slots. Requires addressing overhead in each slot, but maximizes throughput.

AKTU Exam Question: "Discuss different physical layer transmission media" me Guided (Twisted pair, Coax, Fiber) aur Unguided (Radio, Microwave, IR) dono ka neat architectural comparison table zaroor banayein!

Module 13: Switching Techniques: Circuit, Message & Packet

AKTU BCS603 Focus: Circuit vs Message vs Packet switching, store-and-forward latency, pipelining benefits, and Datagram vs Virtual Circuit paradigms.



1. Why Packet Switching Dominates Modern Computer Networks

- Pipelining Effect:** Jab message ko 10 packets me divide kiya jata hai, to Switch 1 packet 2 transmit kar raha hota hai jabki Switch 2 packet 1 ko forward kar raha hota hai. Is concurrent pipelining ke karan overall file transfer delay drastically kam ho jata hai.
- Fault Resilience:** Agar path ka koi router fail ho jaye, to packet switching packets ko dynamically alternate links par detour kar deti hai. Circuit switching me pura call disconnect ho jata hai.

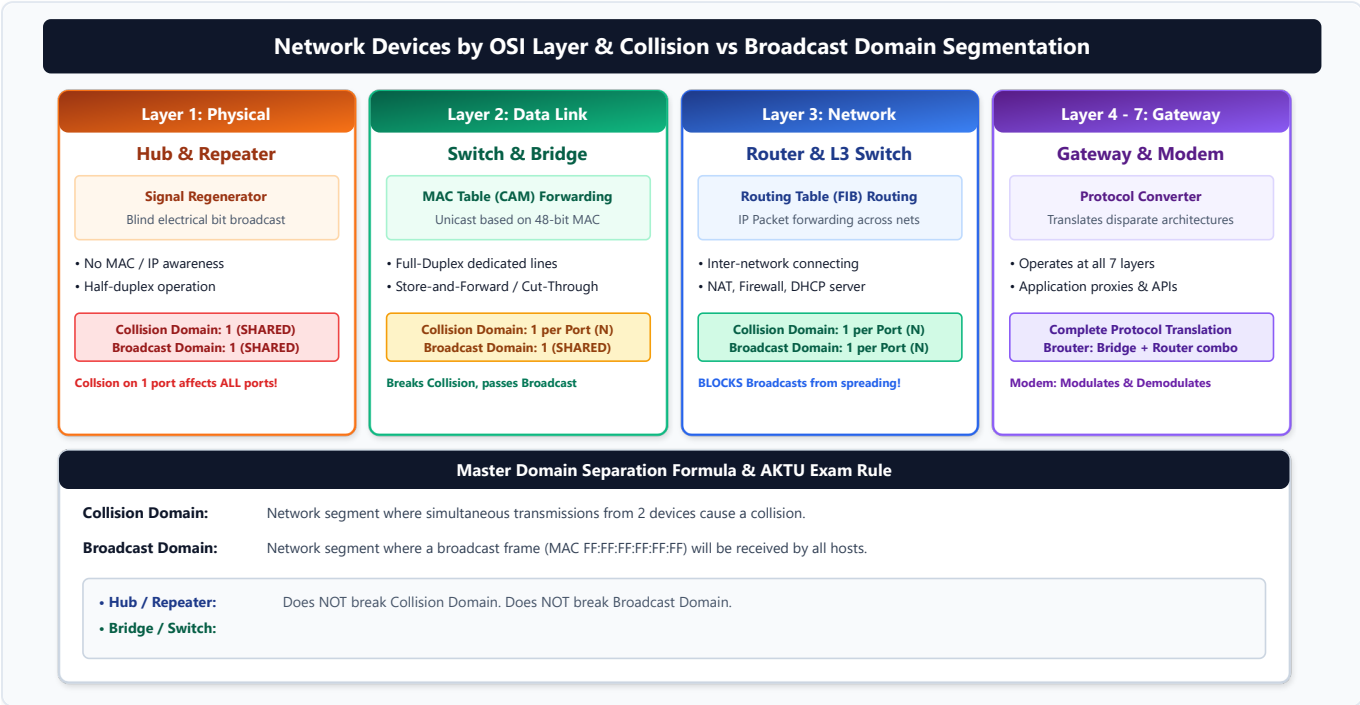
2. Datagram vs Virtual Circuit Approach

Dimension	Datagram Network	Virtual Circuit Network
Connection Setup	Not required (Connectionless)	Mandatory (Setup, Transfer, Teardown)
Packet Addressing	Full Source & Destination Address in each packet	Short Virtual Circuit Identifier (VCI)
Routing Decisions	Made independently for each packet	Made once during call setup for all packets
Packet Sequencing	Can arrive out-of-order	Guaranteed in-order arrival
Router Failure Impact	Packets reroute automatically	Entire circuit terminates

AKTU Exam Question: "Differentiate between circuit switching and packet switching" me 3-phase lifecycle, bandwidth utilization, aur delay curve graph zaroor banayein!

Module 14: Network Devices, Components & Unit 1 AKTU PYQs

AKTU BCS603 Grand Finale: Complete hardware device taxonomy, Collision vs Broadcast domain segmentation rules, and model solutions for AKTU PYQs.



BREAKS Collision Domain (1 domain per port). Does NOT break Broadcast Domain. • Router / Gateway: **BREAKS Collision Domain AND BREAKS Broadcast Domain** (Stops broadcasts cold!).

1. Domain Segmentation Formula (Quick Revision)

Connecting Device	Number of Ports	Collision Domains Created	Broadcast Domains Created
Repeater / Hub	N	1 (Single shared collision pool)	1 (Single broadcast pool)
Bridge / Switch	N	N (1 independent domain per port)	1 (Floods broadcast frames)
Router / Gateway	N	N (1 independent domain per port)	N (Filters/blocks broadcasts)

2. Bridge Learning & Filtering Algorithm

Bridge incoming frames ke source MAC address se network topology learn karta hai:

Frame Arrives on Port P with [Src: MAC_S, Dst: MAC_D]

- Update Table: MAC_S -> Port P
- If MAC_D is in Table:

If Port(MAC_D) == P: DROP / FILTER Frame (Local traffic)

Else: FORWARD Frame to Port(MAC_D)
- Else: FLOOD Frame to all ports except P

AKTU 10-Mark Success Blueprint: Hardware devices question me har device ka OSI layer, working principle (CAM table / Routing table), aur Collision/Broadcast domain separation status zaroor mention karein!